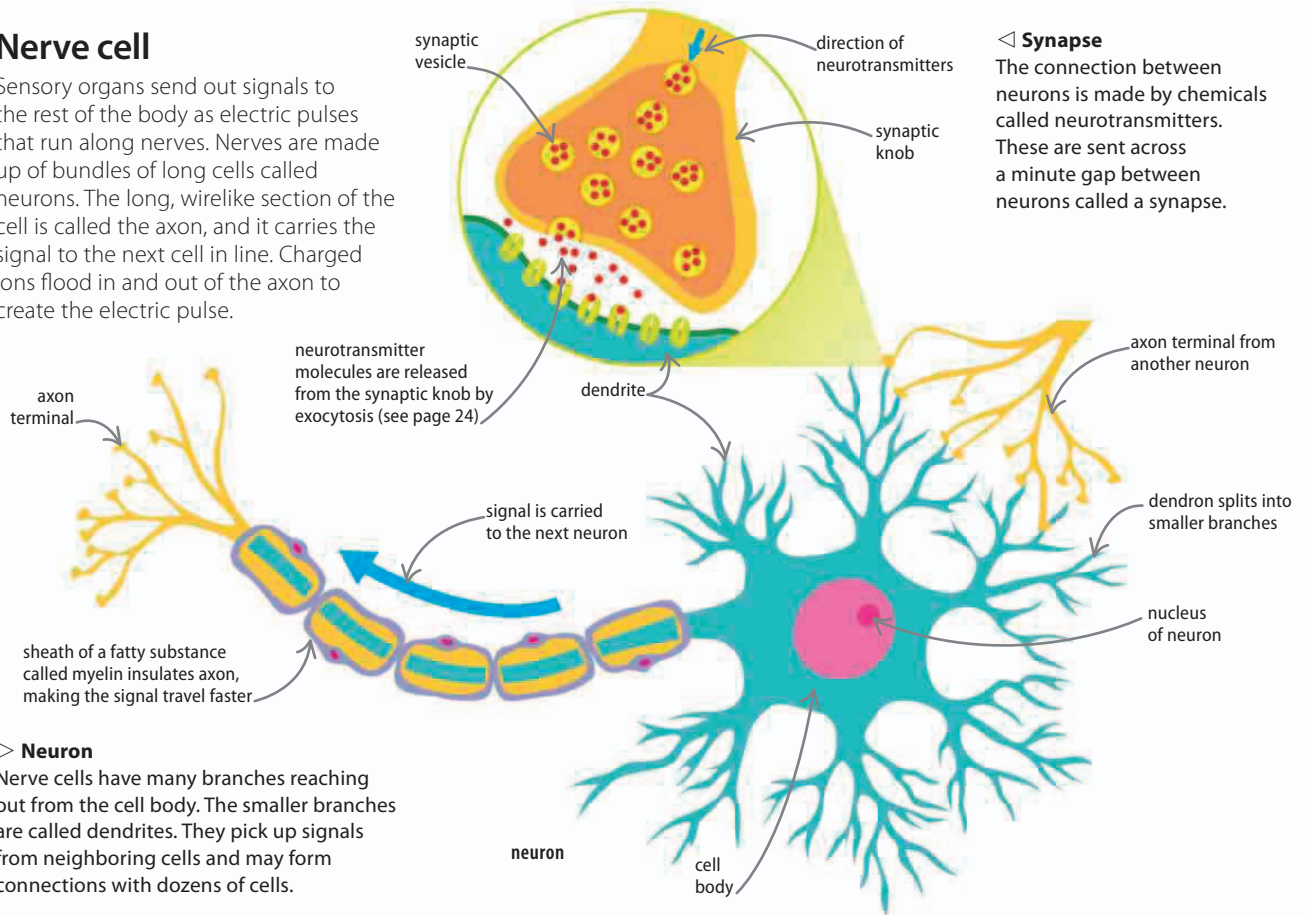


Nerve cell

Sensory organs send out signals to the rest of the body as electric pulses that run along nerves. Nerves are made up of bundles of long cells called neurons. The long, wirelike section of the cell is called the axon, and it carries the signal to the next cell in line. Charged ions flood in and out of the axon to create the electric pulse.



◁ Synapse

The connection between neurons is made by chemicals called neurotransmitters. These are sent across a minute gap between neurons called a synapse.

▷ Neuron

Nerve cells have many branches reaching out from the cell body. The smaller branches are called dendrites. They pick up signals from neighboring cells and may form connections with dozens of cells.

Reflex action

Information from the senses travels toward the brain through sensory neurons. In vertebrates, such as humans, these connect to the spinal cord, and the signal travels up to the brain through the cord. Any immediate response to the stimulus (such as a sharp pin) is sent out to the muscles by motor neurons right away. This means that reflex actions, such as withdrawing the hand from the source of pain, do not involve the brain, but are controlled by the spinal cord alone.

▷ Reflex arc

The nerve pathway controlling a reflex is called the reflex arc. The sensory nerve sends a signal to the spinal cord, where it connects directly to the motor neuron that signals to the muscles, causing them to move.

